EXPLORERS CHOICE · LOCAL DEVELOPMENT

Create the MySQL database on your Windows PC

A practical setup guide for a local MySQL 8 database used by the Explorers Choice backend.

Next.js website → FastAPI backend → MySQL 8 on localhost

What you will set up

- A local MySQL schema named `explorers_choice`.
- A dedicated MySQL account for the app (not the MySQL root account).
- The app's existing MySQL tables, installed with its Alembic migration.

> **Use the project migration, not a hand-written schema.** It creates the exact tables and indexes expected by the backend. The current MySQL migration defines 19 application tables.

Before you begin

1. Install MySQL Server 8.x for Windows and start the MySQL service.
2. Install MySQL Workbench, or use the MySQL command-line client.
3. Open the Explorers Choice project in VS Code. Its root Python environment should be `.venv`.

This guide is for a local MySQL server at localhost:3306, not TiDB Cloud. If MySQL is installed on a different port, use that same port in both MySQL and the backend settings.

1 · Create the database and application user

Connect to your local MySQL server as an administrator in Workbench, open a SQL tab, and run this script. Replace the sample password with a strong password of your own.

```
CREATE DATABASE explorers_choice
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

CREATE USER 'explorers_app'@'localhost'
  IDENTIFIED BY 'REPLACE_WITH_A_STRONG_PASSWORD';

GRANT ALL PRIVILEGES ON explorers_choice.*
  TO 'explorers_app'@'localhost';

FLUSH PRIVILEGES;
```

> **Keep credentials private.** Do not use the MySQL root account for the app and do not paste passwords into source code, screenshots, or chat. Grant access only to `explorers_choice.*`.

CONFIGURATION

2 · Point the backend at local MySQL

Open `backend\.env` in the project. Add or update these database settings, using the same password you chose in the SQL step.

```
DATABASE_PROVIDER=mysql
MYSQL_HOST=localhost
MYSQL_PORT=3306
MYSQL_DATABASE=explorers_choice
MYSQL_USER=explorers_app
MYSQL_PASSWORD=REPLACE_WITH_A_STRONG_PASSWORD
MYSQL_SSL_MODE=none
```

For local MySQL, `MYSQL_SSL_MODE=none` is appropriate. Remote production databases should use TLS verification instead.

Check the application secrets too

The backend also requires a non-placeholder `SECRET_KEY` and `ADMIN_API_KEY`. Set private development values in `backend\.env` if they are not already configured. Never share or commit this file.

> **Important:** Do not replace the entire `backend\.env` with the snippet above if it contains other settings. Edit only the MySQL keys and preserve your existing private configuration.

3 · Install the schema with the migration

Open a PowerShell terminal in VS Code and change to the backend directory:

```
cd backend
```

Run the MySQL migration using the project's root virtual environment:

```
..\.venv\Scripts\python.exe -m alembic -c alembic_mysql.ini upgrade head
```

A successful run creates the application tables and Alembic's migration tracking table. Migrations are not applied automatically when the backend starts.

> **Use the MySQL configuration file.** The command must include `-c alembic_mysql.ini`. Do not use the default PostgreSQL migration configuration for this local MySQL database.

What the database contains

Accounts and reset tokens; destinations, packages, itineraries and FAQs; package bookings, travellers, payments, documents and notes; enquiries, hotels, train and cab bookings; offers, customer stories, settings and audit logs.

RUN AND VERIFY

4 · Confirm the tables exist

In MySQL Workbench, run:

```
USE explorers_choice;
SHOW TABLES;
SELECT version_num FROM alembic_version;
```

You should see 19 application tables, including `users`, `destinations`, `packages`, `bookings`, `train_bookings`, and `cab_bookings`, plus `alembic_version`.

5 · Start the backend

From the project root, run:

```
cd backend
..\.venv\Scripts\python.exe -m uvicorn app.main:app --reload --port 8000
```

When it starts, open `http://localhost:8000/docs` in your browser. Then, in a second terminal at the project root, start the website:

```
npm run dev
```

If the connection fails

Message or symptom	Check
Connection refused / can't connect	MySQL Server service is running; host and port are correct.
Access denied	Username/password match; the user has privileges on `explorers_choice.*`.
Unknown database	Schema exists and `MYSQL_DATABASE=explorers_choice`.
Missing settings / placeholder secret	Backend `.env` has the MySQL values and non-placeholder app secrets.
Migration command cannot find Python	Run it from `backend`; confirm root `.venv\Scripts\python.exe` exists.

> **Do not delete the database to fix a connection error.** Fix the connection settings first. Back up any database with data before running schema changes.

Project references: `backend\app\models.py`, `backend\app\migrations_mysql\versions\0001_initial_mysql_schema.py`, and `backend\app\config.py`.